

BANKI NAVYA(2520090189)

## SECTION 1: DDL

### Q1 – CREATE TABLE

```
MySQL 8.0 Command Line Client
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 21
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE bank;
Query OK, 1 row affected (0.01 sec)

mysql> USE bankdb;
Database changed
mysql> CREATE TABLE bank_transactions (
  ->   txn_id INT PRIMARY KEY,
  ->   customer_name VARCHAR(50),
  ->   branch_name VARCHAR(50),
  ->   transaction_type VARCHAR(20),
  ->   amount DECIMAL(10,2),
  ->   transaction_date DATE
  -> );
ERROR 1050 (42S01): Table 'bank_transactions' already exists
mysql> SHOW TABLES;
+-----+
| Tables_in_bankdb |
+-----+
| bank_transactions |
+-----+
1 row in set (0.04 sec)

mysql> DESCRIBE bank_transactions;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| txn_id         | int           | NO   | PRI | NULL    |       |
| customer_name  | varchar(50)   | YES  |     | NULL    |       |
| branch_name    | varchar(50)   | YES  |     | NULL    |       |
| transaction_type | varchar(20)   | YES  |     | NULL    |       |
| amount         | decimal(10,2) | YES  |     | NULL    |       |
| transaction_date | date          | YES  |     | NULL    |       |
| account_no     | varchar(20)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
7 rows in set (0.01 sec)
```

## Q2 – ALTER TABLE ADD COLUMN

```
mysql> ALTER TABLE bank_transactions
-> ADD account_no VARCHAR(20);
ERROR 1060 (42S21): Duplicate column name 'account_no'
mysql> ALTER TABLE bank_transactions
-> ADD account_no VARCHAR(20);
ERROR 1060 (42S21): Duplicate column name 'account_no'
mysql> DESCRIBE bank_transactions;
```

| Field            | Type          | Null | Key | Default | Extra |
|------------------|---------------|------|-----|---------|-------|
| txn_id           | int           | NO   | PRI | NULL    |       |
| customer_name    | varchar(50)   | YES  |     | NULL    |       |
| branch_name      | varchar(50)   | YES  |     | NULL    |       |
| transaction_type | varchar(20)   | YES  |     | NULL    |       |
| amount           | decimal(10,2) | YES  |     | NULL    |       |
| transaction_date | date          | YES  |     | NULL    |       |
| account_no       | varchar(20)   | YES  |     | NULL    |       |

```
7 rows in set (0.00 sec)
```

## Q3 – ALTER TABLE MODIFY COLUMN

```
mysql> ALTER TABLE bank_transactions
-> MODIFY customer_name VARCHAR(100);
Query OK, 10 rows affected (0.17 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql>
```

## Q4 – RENAME TABLE

```
mysql> RENAME TABLE bank_transactions TO customer_transactions;
Query OK, 0 rows affected (0.03 sec)

mysql>
```

## Q5 – TRUNCATE TABLE

```
mysql> TRUNCATE TABLE customer_transactions;
Query OK, 0 rows affected (0.06 sec)

mysql>
```

## Q6 – DROP TABLE

```
mysql> CREATE TABLE bank_backup AS  
-> SELECT * FROM customer_transactions;  
Query OK, 0 rows affected (0.04 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

## Conclusion of section 1

```
mysql> SHOW TABLES;
```

```
+-----+  
| Tables_in_bankdb |  
+-----+  
| bank_backup      |  
| customer_transactions |  
+-----+  
2 rows in set (0.00 sec)
```

```
mysql> DESCRIBE customer_transactions;
```

```
+-----+-----+-----+-----+-----+-----+  
| Field          | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| txn_id         | int           | NO   | PRI | NULL    |       |  
| customer_name  | varchar(100)  | YES  |     | NULL    |       |  
| branch_name    | varchar(50)   | YES  |     | NULL    |       |  
| transaction_type | varchar(20)   | YES  |     | NULL    |       |  
| amount         | decimal(10,2) | YES  |     | NULL    |       |  
| transaction_date | date          | YES  |     | NULL    |       |  
| account_no     | varchar(20)   | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
7 rows in set (0.02 sec)
```

## SECTION - 2

### Q7 – INSERT

```
mysql> INSERT INTO customer_transactions
  -> (txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
  -> VALUES
  -> (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'),
  -> (102,'Sita','Hyderabad','Withdrawal',2000,'2024-01-06'),
  -> (103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08'),
  -> (104,'Anil','Vizag','Deposit',8000,'2024-01-10'),
  -> (105,'Priya','Hyderabad','Withdrawal',3500,'2024-01-11'),
  -> (106,'Ramesh','Vizag','Deposit',15000,'2024-01-12'),
  -> (107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13'),
  -> (108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14'),
  -> (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15'),
  -> (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');
Query OK, 10 rows affected (0.03 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> _
```

### Q8 – UPDATE

```
mysql> UPDATE customer_transactions
  -> SET amount = 5000
  -> WHERE txn_id = 105;
Query OK, 1 row affected (0.09 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> _
```

\*\*\*\*\*

```
mysql> RENAME TABLE customer_transactions TO bank_transactions;
Query OK, 0 rows affected (0.05 sec)

mysql> _
```

### Q9 – DELETE

```
mysql> ^C
mysql> DELETE FROM bank_transactions
    -> WHERE txn_id = 111;
Query OK, 0 rows affected (0.01 sec)

mysql>
```

## Q10 – SELECT

```
mysql> SELECT *
    -> FROM bank_transactions;
```

| txn_id | customer_name | branch_name | transaction_type | amount   | transaction_date | account_no |
|--------|---------------|-------------|------------------|----------|------------------|------------|
| 101    | Ravi          | Hyderabad   | Deposit          | 5000.00  | 2024-01-05       | NULL       |
| 102    | Sita          | Hyderabad   | Withdrawal       | 2000.00  | 2024-01-06       | NULL       |
| 103    | Kiran         | Vijayawada  | Deposit          | 12000.00 | 2024-01-08       | NULL       |
| 104    | Anil          | Vizag       | Deposit          | 8000.00  | 2024-01-10       | NULL       |
| 105    | Priya         | Hyderabad   | Withdrawal       | 5000.00  | 2024-01-11       | NULL       |
| 106    | Ramesh        | Vizag       | Deposit          | 15000.00 | 2024-01-12       | NULL       |
| 107    | Keerthi       | Vijayawada  | Withdrawal       | 1000.00  | 2024-01-13       | NULL       |
| 108    | Rahul         | Hyderabad   | Deposit          | 9000.00  | 2024-01-14       | NULL       |
| 109    | Sneha         | Vizag       | Withdrawal       | 4000.00  | 2024-01-15       | NULL       |
| 110    | Madhu         | Vijayawada  | Deposit          | 11000.00 | 2024-01-16       | NULL       |

```
10 rows in set (0.00 sec)

mysql>
```

## Q11 – SELECT WITH WHERE

```
mysql> SELECT *
    -> FROM bank_transactions
    -> WHERE transaction_type='Deposit';
```

| txn_id | customer_name | branch_name | transaction_type | amount   | transaction_date | account_no |
|--------|---------------|-------------|------------------|----------|------------------|------------|
| 101    | Ravi          | Hyderabad   | Deposit          | 5000.00  | 2024-01-05       | NULL       |
| 103    | Kiran         | Vijayawada  | Deposit          | 12000.00 | 2024-01-08       | NULL       |
| 104    | Anil          | Vizag       | Deposit          | 8000.00  | 2024-01-10       | NULL       |
| 106    | Ramesh        | Vizag       | Deposit          | 15000.00 | 2024-01-12       | NULL       |
| 108    | Rahul         | Hyderabad   | Deposit          | 9000.00  | 2024-01-14       | NULL       |
| 110    | Madhu         | Vijayawada  | Deposit          | 11000.00 | 2024-01-16       | NULL       |

```
6 rows in set (0.00 sec)

mysql>
```

## Q12 – SELECT WITH ORDER BY

```
mysql> SELECT *
-> FROM bank_transactions
-> ORDER BY amount DESC;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 | NULL |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 | NULL |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 | 2024-01-16 | NULL |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 | 2024-01-14 | NULL |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 | NULL |
| 101 | Ravi | Hyderabad | Deposit | 5000.00 | 2024-01-05 | NULL |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 | NULL |
| 109 | Sneha | Vizag | Withdrawal | 4000.00 | 2024-01-15 | NULL |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 | NULL |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00 | 2024-01-13 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql>
```

## SECTION 3

### Q13 – GRANT SELECT

```
mysql> GRANT SELECT
-> ON bank_transactions
-> TO Auditor1;
ERROR 1410 (42000): You are not allowed to create a user with GRANT
mysql> GRANT SELECT
-> ON bank_transactions
-> TO Auditor1;
ERROR 1410 (42000): You are not allowed to create a user with GRANT
mysql> CREATE USER 'Auditor1'@'localhost' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.06 sec)

mysql> GRANT SELECT
-> ON bank_transactions
-> TO Auditor1;
ERROR 1410 (42000): You are not allowed to create a user with GRANT
mysql> GRANT SELECT
-> ON bankdb.bank_transactions
-> TO 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql>
```

### Q14 – GRANT ALL PRIVILEGES

```
mysql> GRANT ALL
-> ON bank_transactions
-> TO BranchManager;
ERROR 1410 (42000): You are not allowed to create a user with GRANT
mysql> CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY 'password123';
Query OK, 0 rows affected (0.02 sec)

mysql> GRANT ALL PRIVILEGES
-> ON bankdb.bank_transactions
-> TO 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.02 sec)

mysql>
```

## Q15 – REVOKE SELECT

```
mysql> REVOKE SELECT
  -> ON bank_transactions
  -> FROM Auditor1;
ERROR 1147 (42000): There is no such grant defined for user 'Auditor1' on host '%' on table 'bank_transactions'
mysql> REVOKE SELECT
  -> ON bankdb.bank_transactions
  -> FROM 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.02 sec)

mysql>
```

## Q16 – REVOKE ALL

```
mysql> REVOKE ALL
  -> ON bank_transactions
  -> FROM BranchManager
  -> REVOKE ALL
  -> ON bank_transactions
  -> FROM BranchManager;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server
version for the right syntax to use near 'REVOKE ALL
ON bank_transactions
FROM BranchManager' at line 4
mysql> REVOKE ALL PRIVILEGES
  -> ON bankdb.bank_transactions
  -> FROM 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.01 sec)
```

## SECTION 4

## Q17 – COMMIT

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE bank_transactions
  -> SET amount = 6000
  -> WHERE txn_id = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql>
mysql> COMMIT;
```

## Q18 – SAVEPOINT

```
mysql>
mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.00 sec)

mysql>
```

## Q19 – ROLLBACK TO SAVEPOINT

```
mysql> UPDATE bank_transactions
  -> SET amount = 99999
  -> WHERE txn_id = 102;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> SELECT * FROM bank_transactions
  -> WHERE txn_id = 102;
+-----+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date | account_no |
+-----+-----+-----+-----+-----+-----+-----+
| 102 | Sita | Hyderabad | Withdrawal | 99999.00 | 2024-01-06 | NULL |
+-----+-----+-----+-----+-----+-----+-----+
1 row in set (0.01 sec)
```

## Q20 – FULL TCL EXAMPLE

```
mysql>
mysql> SAVEPOINT SP1;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> UPDATE bank_transactions
  -> SET amount = 9000
  -> WHERE txn_id = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
```